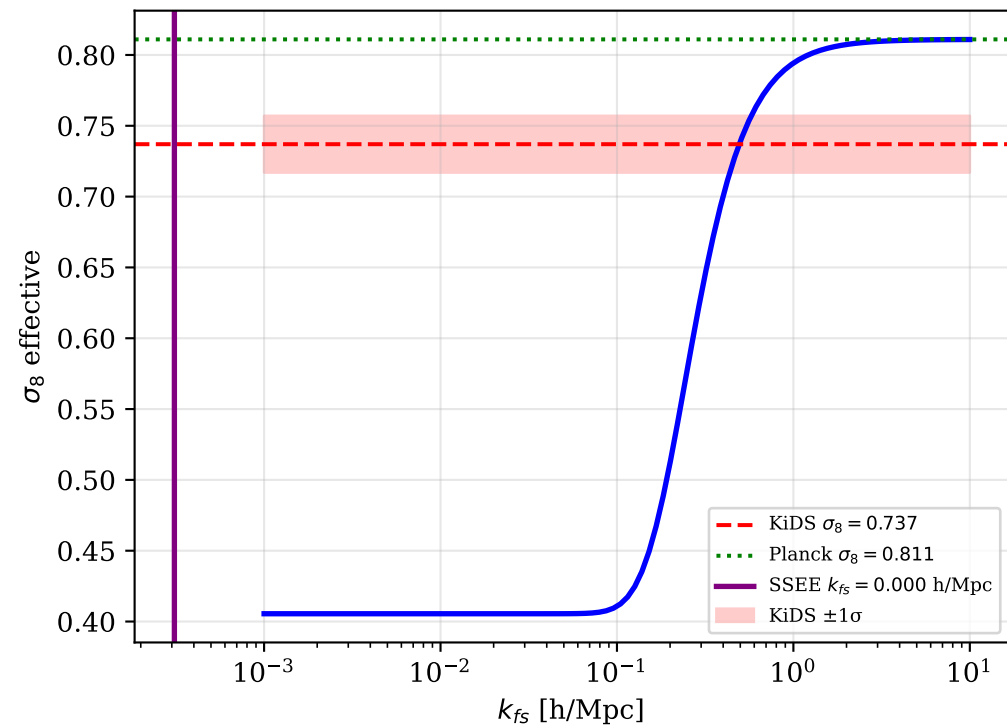
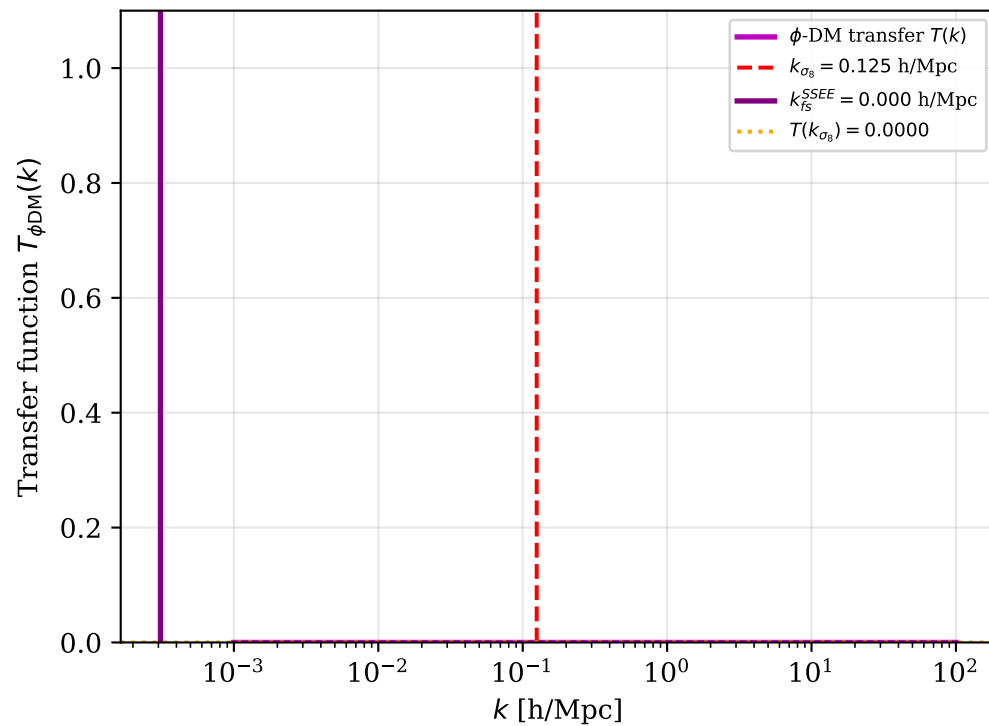


Two-Sector DM  
 $\sigma_8$  vs free-streaming scale



$\phi$ -DM Suppression



$\phi$ -DM Decoupling Timeline  
 $(\tau_{\text{I}}H = 1$  condition)

